

ChristosHadjinikolis

Engineering Manager | Production ML & Streaming Systems

Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21

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mediums



organisation

org scale-up
hiring loops
mentoring/retention
cross-functional delivery

systems

client-facing ML
high-volume streaming
observability
reliability

engineering

operating models
streaming platforms
architecture decisions
local E2E tests

technical

ML systems
Java/Kotlin
Python/SQL
Kafka/Flink
Airflow/AWS/GCP

public

JTC'21 standards
UCL research
PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

Engineering Manager for ML and real-time data systems with **16 years** across AI research, consulting, backend engineering, and live data platforms. Lead teams that turn noisy data, experimental models, and ambiguous product semantics into reliable, client-facing systems. At Vortexa, own a major **live streaming backend estate**, lead a **10-person cross-functional team**, and drove Data Production Team growth from **4 to 30+ people** by shaping hiring, system-design culture, platform choices, and operating discipline. My UCL and JTC'21 standards work informs how I treat auditability, explainability, versioning, and safe adoption in practice. I build engineering organisations, operating models, and platforms that keep ML systems dependable when product quality and client trust are on the line.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own the **engineering strategy and delivery** for Vortexa's **live streaming intelligence estate**, a major client-facing backend area powering product and commercial workflows, filtering roughly **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Lead **live-processing strategy** with a **10-person cross-functional team** spanning **6 MLE/DE/DS, 2 Product, and 2 SMEs**. Because this estate feeds live product behaviour, reliability, explainability, monitoring, and response readiness are delivery requirements.

- **People & Org Scaling:** Directly manage the MLE/DE/DS core and lead a **10-person cross-functional delivery team**; drove Data Production Team growth from **4 to 30+** by developing hiring loops and system design evaluations, hired **5 senior DS/DEs**, evaluated **31 candidates**, retained/progressed the full team, and reduced key-person risk.
- **Platform Scaling:** Enabled independent scaling of critical live pipelines by leading the two-year **Kafka Streams-to-Flink migration**; used Flink's dataflow model, independent state/checkpointing, richer operational UI, and framework-driven design to reduce maintenance effort.
- **Trustworthy ML Estate:** Established Vortexa's ML development lifecycle from research to production: prototyping discipline, **MLflow**, model/data versioning, service-based model deployment, replay/evaluation interfaces, and live accuracy-monitoring dashboards.
- **Reliability & Ops Culture:** Protect customer-facing reliability with monitoring, runbooks, rollback/fallback plans, and Jira alert workflows; moved from firefighter patterns to shared on-call ownership while keeping **MTTR under 30 min**.
- **Product Semantics & Interfaces:** Resolve recurring prediction-quality disputes through shared semantics, turning stakeholder friction into measurable evaluation targets and RFC/workstream-ready product interfaces.
- **Engineering Operating Model:** Scale delivery through Flink/Airflow archetypes, ADRs, docs close to code, tests-as-docs, dev containers, Backstage adoption, local E2E/integration tests, and a weekly architecture forum; cut a **3h DAG feedback loop to under 5 min** and reduced single-owner review bottlenecks.

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and played a formative role in growing it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; led client meetings, scoped goals, placed consultants, interviewed/mentored hires, and delivered ML/data work across UBS, Vodafone, CNHi and other enterprise clients. Ran the London Flink Meetup for five years.

- **Client Delivery Leadership:** Drove agendas with client stakeholders and product teams, agreed goals, facilitated technical delivery, and supported hiring/consultant placement across banking, telecoms, manufacturing, publishing, and gaming accounts.
- **Vodafone Group:** Led technical delivery across multiple workstreams; worked on Infinity, a GCP/**Kubeflow** DS platform, shaping ML lifecycle support; built Red Agent, a mobile-network feature-engineering framework.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance; managed/guided a DS team using **PySpark** over live vehicle sensory data with product-facing backlog discipline.
- **UBS:** Grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability work, and real-time pipelines with **Kafka**, **Elasticsearch**, and Python.

Roles in UK Institutions 2010–2016

Coding Teacher @DGC | Teaching Assistant @KCL/UCL

Built the communication and presentation discipline I now rely on as a leader by teaching Java/Python and core computer science at David Game College from 09/2012 to 04/2016 while completing doctoral research; Teaching Assistant at KCL and UCL.

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning, graph analytics , and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics. Received Best Poster Award at IJCAI 2013 (413 submissions). Awarded EPSRC PhD Scholarship , Outstanding TA Award (student-nominated), and earned the Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering with a strong focus on mathematics (10 courses), covering theoretical aspects, networking, and telecommunications. Majored in Artificial Intelligence .	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

committees/affiliations

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa , Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio , a published PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science explained by an ML expert, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

certificates & awards

07/2020	AWS Certified Machine Learning - Specialty Validation Number: D0R0WC7CN1VQ13SV	AWS
01/2019	Process Mining License number: AFU2VTP6273V	Coursera Course Certificates
10/2017	Google Certified Professional - Data Engineer Credential ID: 000000400	Google Cloud Certified
07/2017	Advanced Elasticsearch: Data modelling Enrolment Id: 33442	Elastic
06/2016	Graph Analytics for Big Data License number: NZADYQA7N8FW	Coursera Course Certificates
05/2016	Apache Spark Developer Certification License number: 1.x-0548	O'Reilly Media
04/2016	Neo4j Certified Professional License number: here	Neo Technology, Creators of Neo4j